

Supplementary Information

Expression-to-chromatin decoding of the *Arabidopsis* spaceflight transcriptome

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Supplementary tables

Tables S1–S57 and D1–D7 are CSV files in `supplementary_tables/` and `data/derived/` of the archive. The index there (`supplementary_tables/README.md`) gives, for each table, its content, the script that writes it, the manuscript section that cites it, and the audit notes on how to read it.

Supplementary figures

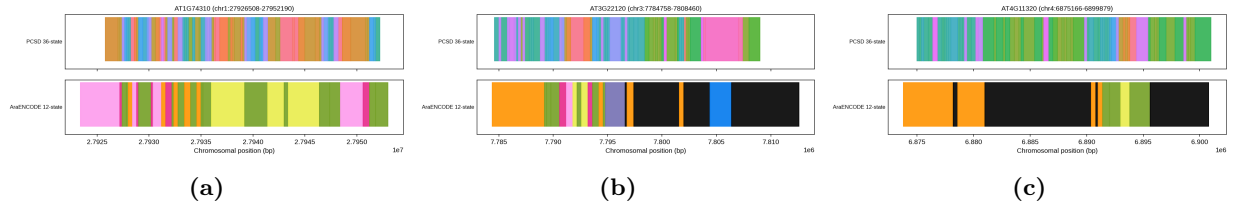


Figure S1: Locus views of three OSD-37 differentially expressed genes, showing PCSD 36-state and AraENCODE 12-state tracks across the gene body and 2 kb upstream (v1).

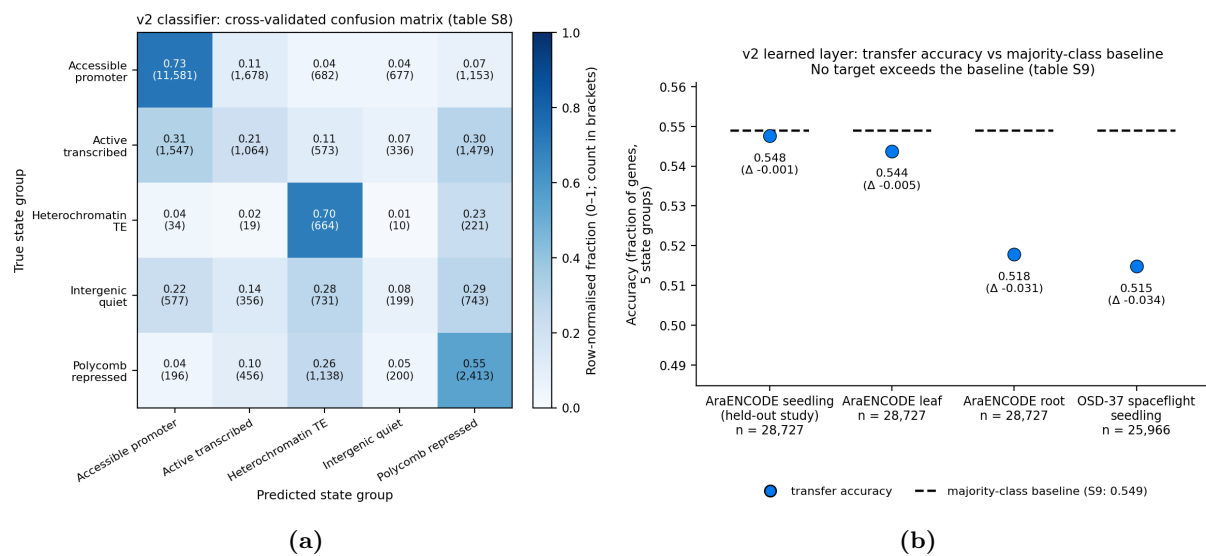


Figure S4: v2 learned layer (tissue-matched AraENCODE training). (a) Confusion matrix (Table S8). (b) Transfer accuracy against the majority-class baseline (Table S9); no transfer exceeds the baseline.

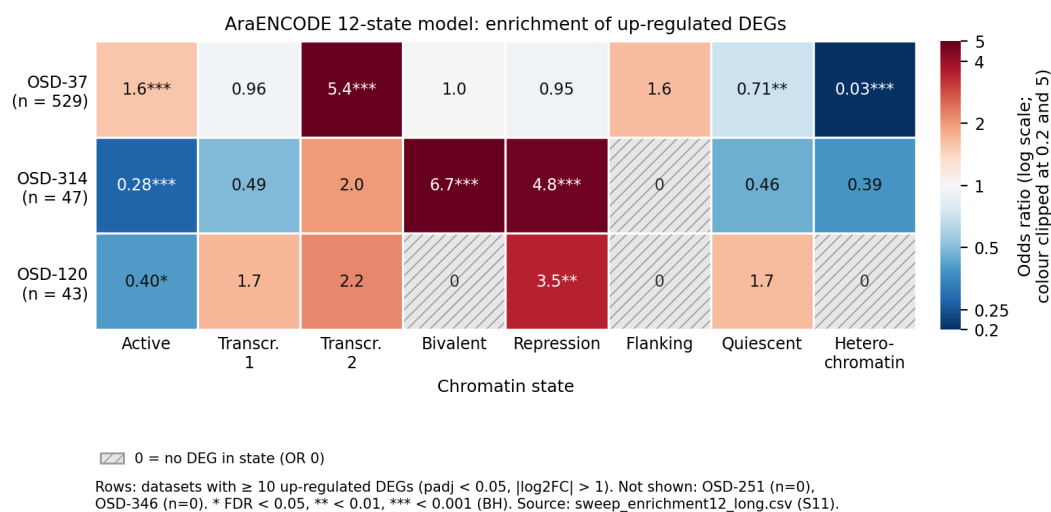
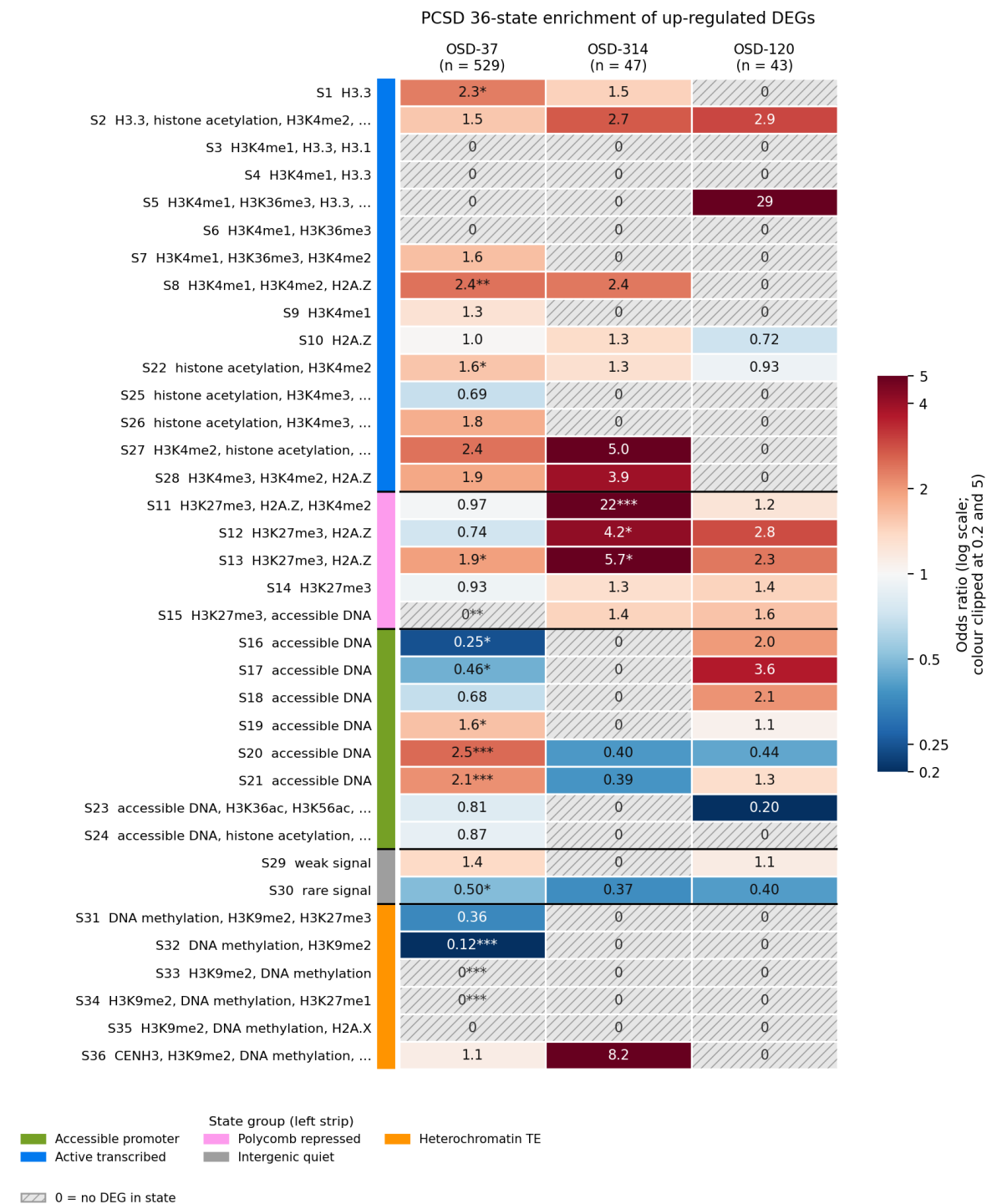


Figure S5: AraENCODE 12-state enrichment of up-regulated DEGs across OSDR datasets (Table S11).



Columns: datasets with ≥ 10 up-regulated DEGs (padj < 0.05, $|\log_2FC| > 1$); not shown: OSD-251 (n=0), OSD-346 (n=0). * FDR < 0.05, ** < 0.01, *** < 0.001 (BH). Mark lists shortened (...); full lists in the PCSD state table. Source: sweep_enrichment36_long.csv (S12).

Figure S6: PCSD 36-state enrichment of up-regulated DEGs across OSDR datasets (Table S12). The OSD-314 row uses the published pooled contrast; corrected values are in main-text Figs. 3 and 7.

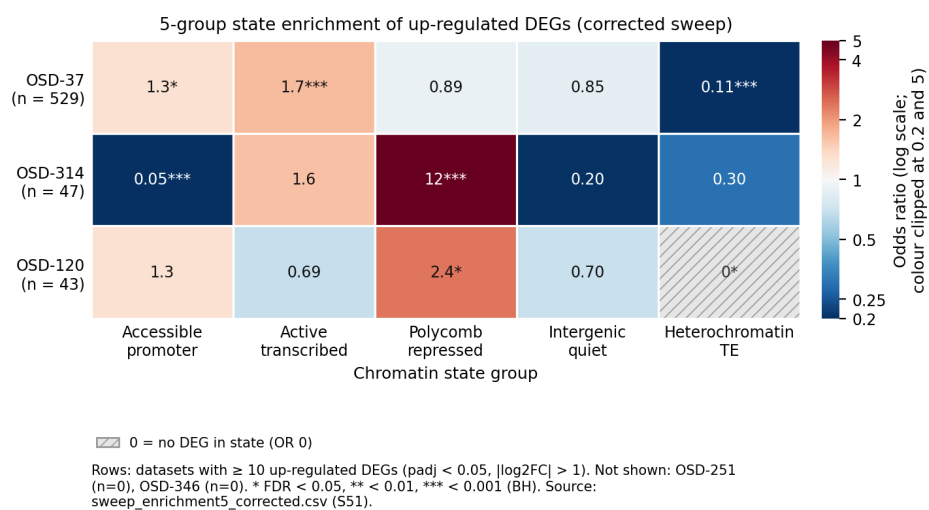


Figure S7: 5-group state enrichment of up-regulated DEGs across OSDR datasets (corrected sweep, Table S51).

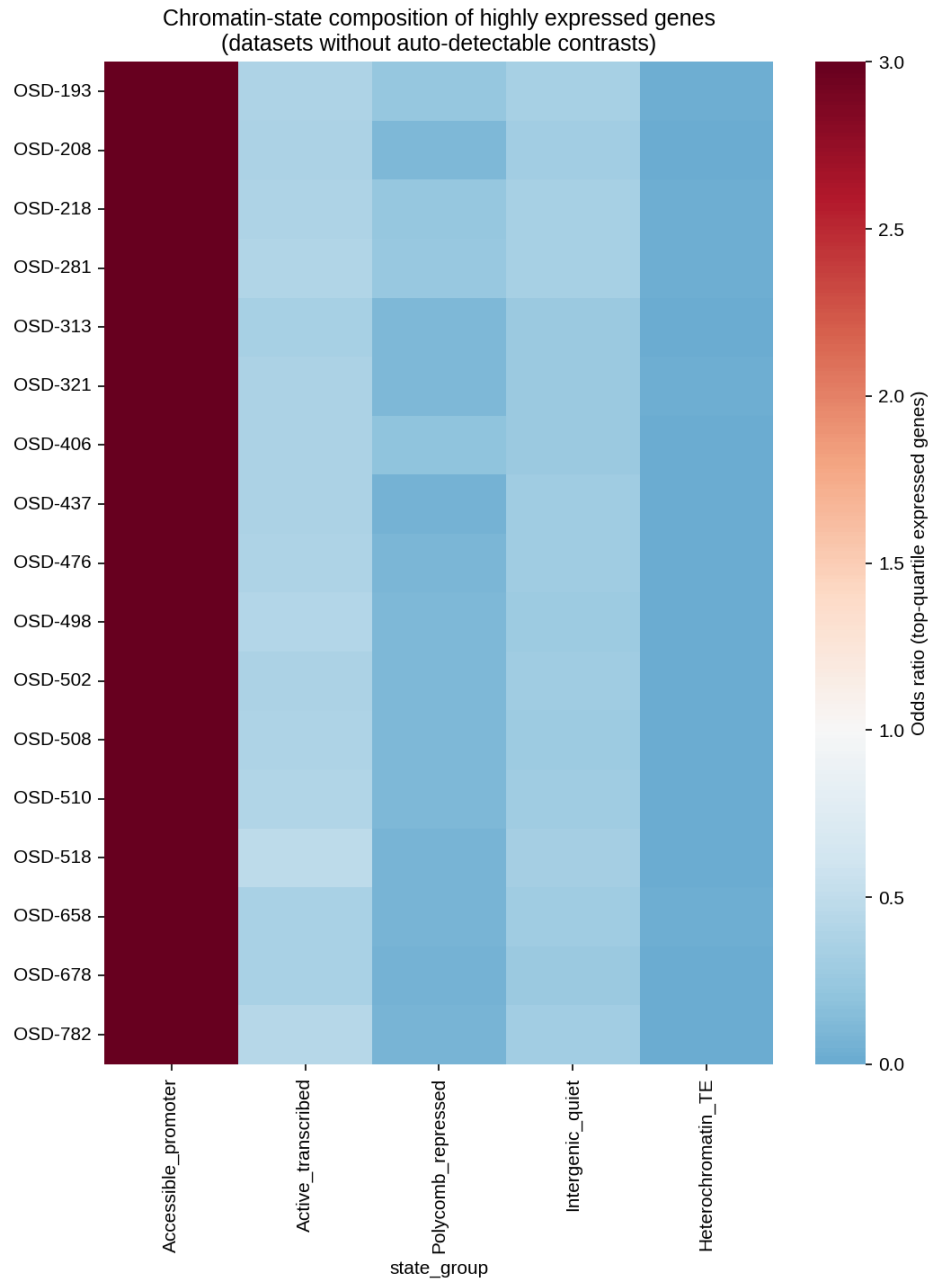


Figure S8: Expression-only fallback decoding across the 18 datasets without a detectable contrast: highly expressed genes sit in accessible-promoter chromatin in every dataset, so the panel is a sanity check and carries no condition-specific signal (Table S10).

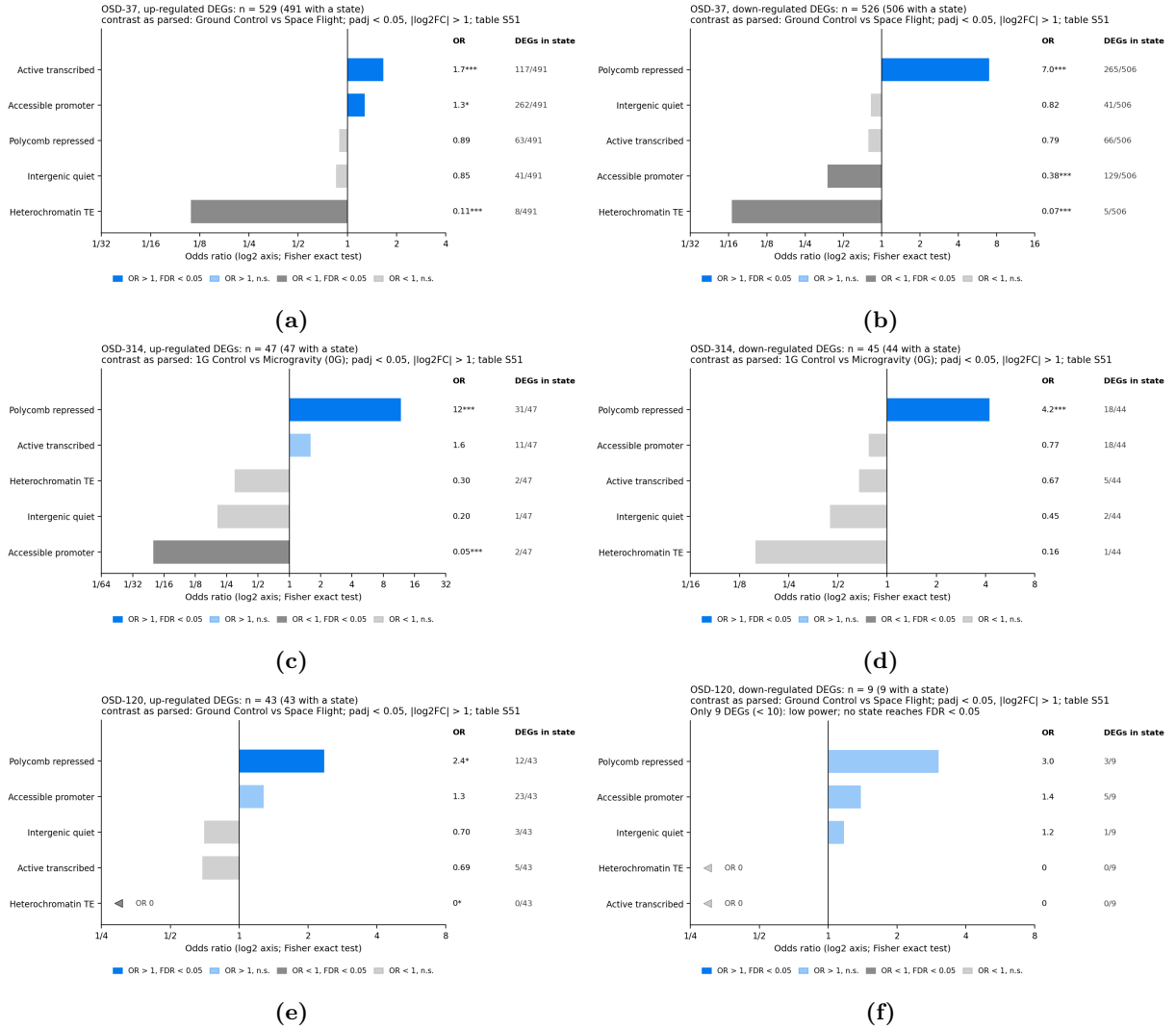


Figure S9: Per-dataset 5-group enrichment of up- and down-regulated DEGs: (a, b) OSD-37, (c, d) OSD-314 (published pooled contrast), (e, f) OSD-120 (mixed genotypes). OSD-251 and OSD-346 have at most one DEG and are not estimable (Table S51).

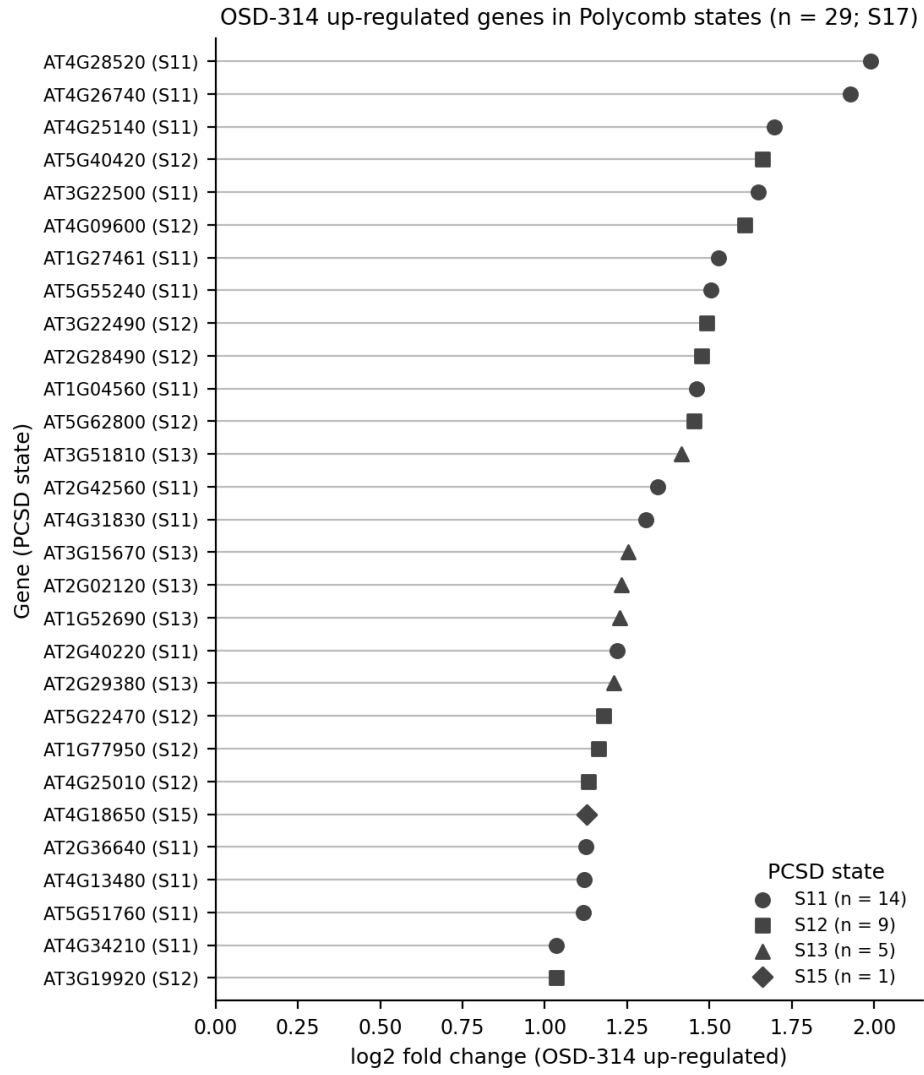


Figure S10: The 29 OSD-314 up-regulated genes in Polycomb states, in the published pooled contrast (Table S17). The corrected contrasts give 255 (0.3 g) and 302 (0 g) such genes (Table S62).

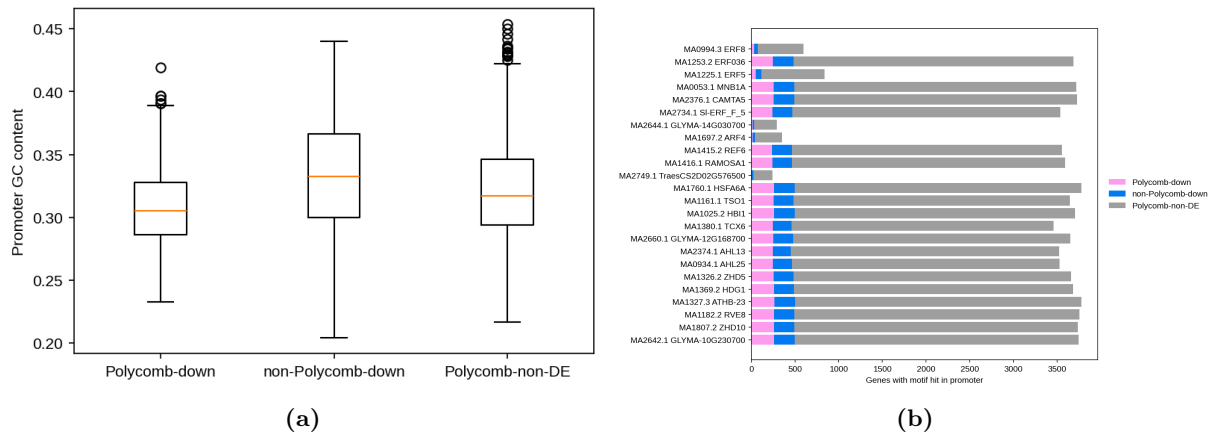


Figure S11: Motif-analysis controls. (a) Mean promoter GC content per gene set (Table S20). (b) Overlap of top-motif hits among Polycomb-down promoters.



Figure S12: The 12 most enriched and 12 most depleted JASPAR motifs in Polycomb-down promoters (Table S18).

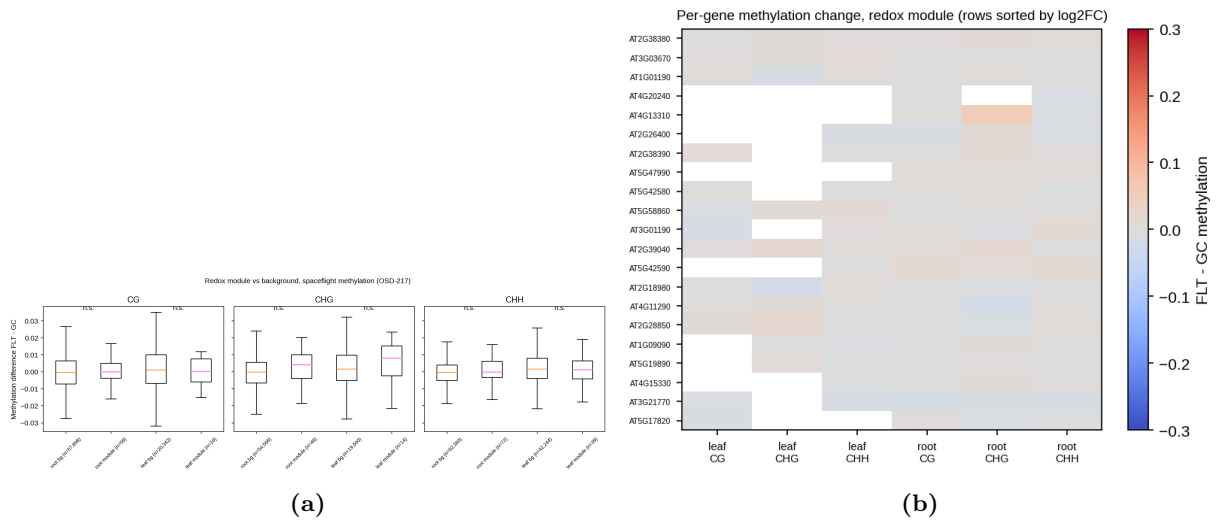


Figure S13: OSD-217 gene-body methylation, flight minus ground, for the redox module vs background (Table S22). (a) Distributions per tissue and context. (b) Per-gene values.

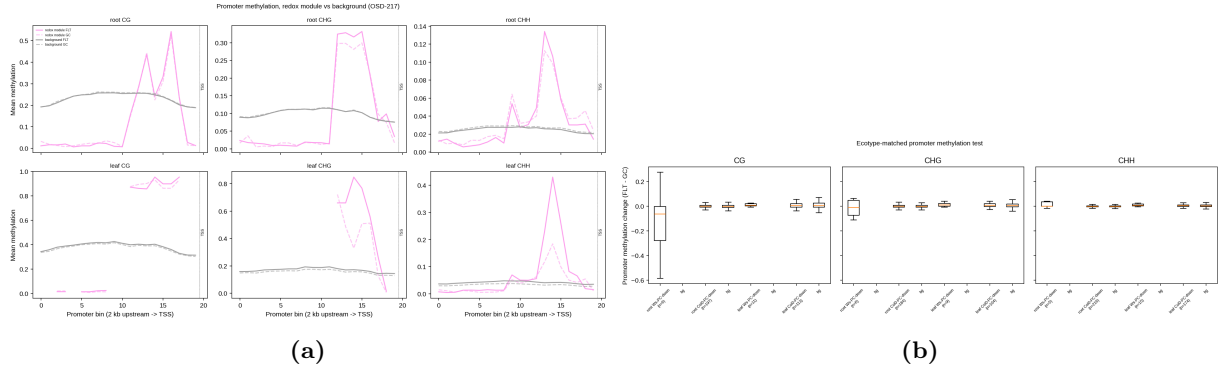


Figure S14: OSD-217 promoter-window methylation (2 kb, 100-bp bins; Tables S23–S24). (a) Metaplots by gene set; leaf coverage is sparse, so the leaf traces are fragmented. (b) Ecotype-matched comparison using the Ws-defined Polycomb-down module.

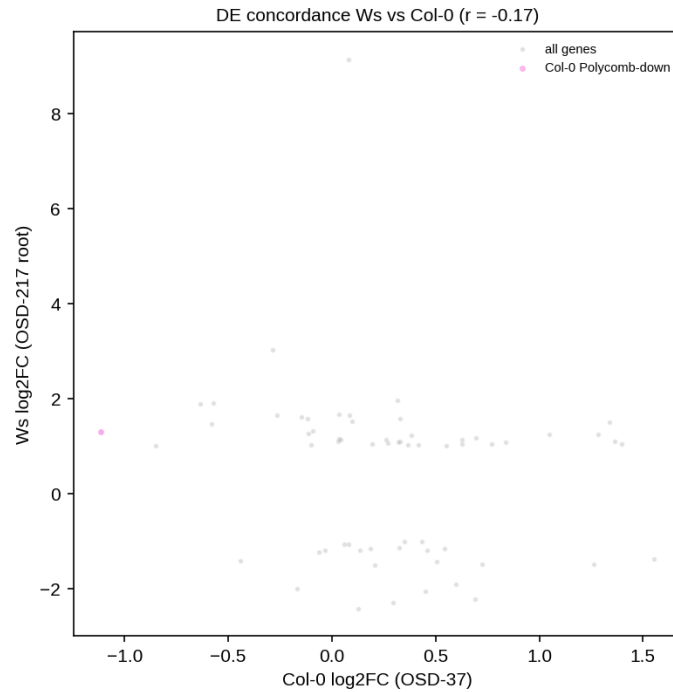


Figure S15: Flight log₂FC concordance between OSD-217 (Ws root) and OSD-37 (Col-0 seedlings), $r = -0.17$ over 63 genes (Table S25).

